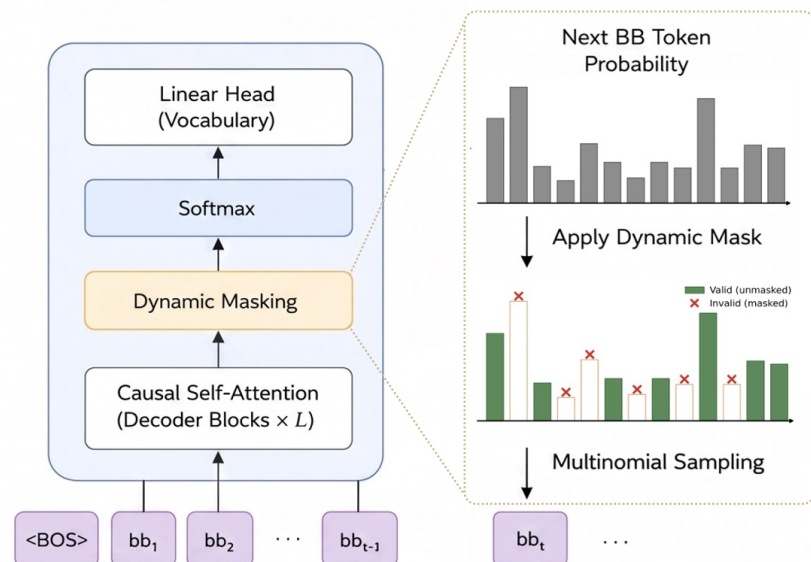


A Generation (Constrained Decoding)

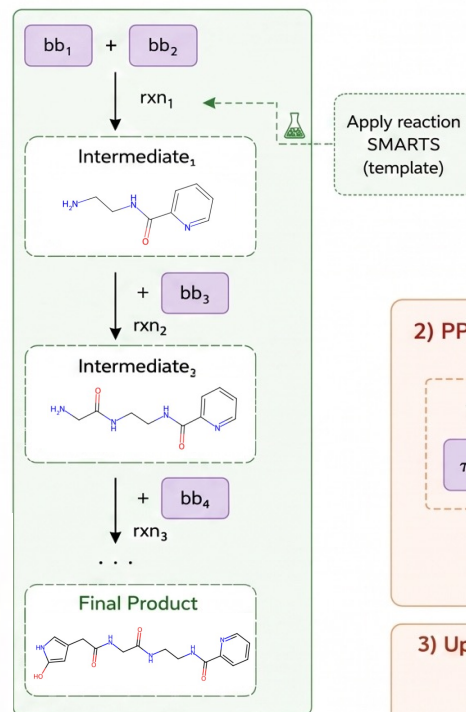
Decoder-only Transformer (GPT-2 style)



RL Loop: Generation $\rightarrow$ Evaluation $\rightarrow$ Update $\rightarrow$ Generation ...

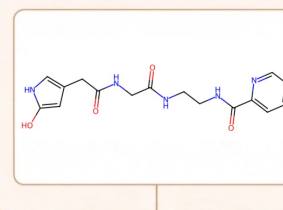
B Sequence-to-Product Conversion

Deterministic Assembly with Curated Reactions



C Evaluation & RL Update

1) Evaluate Final Product



Example Evaluation Signals

- Docking Score
- QSAR / Property
- Synthetic Accessibility
- Novelty / Diversity
- ...

Reward (R)

2) PPO Update (Policy Optimization)

Trajectory Rollouts

$\tau^1 \quad \tau^2 \quad \dots \quad \tau^N$

- Compute rewards
- Estimate advantages (GAE)
- PPO clipped objective

$$\mathcal{L}^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

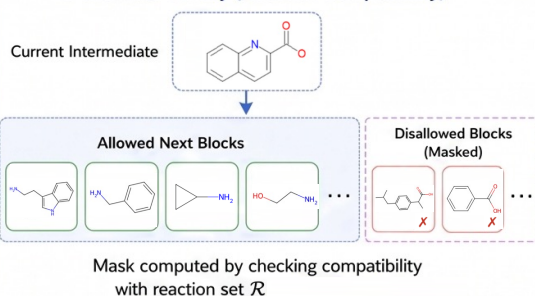
- Update policy parameters θ (using PPO)

3) Update Reference Policy (Optional)

Slowly update reference policy
 $\theta_{\text{ref}} \leftarrow \rho \theta_{\text{ref}} + (1 - \rho) \theta_{\text{old}}$
every K iterations

D Constrained Decoding via Dynamic Masking (Enforces Chemical Validity & Project Constraints)

1. Chemical Validity (Reaction Compatibility)



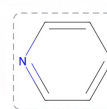
2. Reaction Constraint (Custom Reaction Set)

✓ Preferred — robust, stable linkages	
✓ Amide coupling	<chem>[C:1]=[O:2].[OH].[N:3]>>[C:1]=[O:2].[N:3]</chem>
✓ Suzuki coupling	<chem>[c:1][B+].[c:2][B].[OH].[OH]>>[c:1][c:2]</chem>
✓ Reductive amination	<chem>[C:1]=[O:2].[N:3]>>[C:1][N:3]</chem>
✗ Down-weighted / excluded — unstable linkages	
✗ Ketone bridge	<chem>[cH:1].[C:1][C:2]=[O:3]>>[c:1][C:2]=[O:3]</chem>
✗ C-S / thioether	<chem>[c:1][F].[S:2]>>[c:1][S:2]</chem>

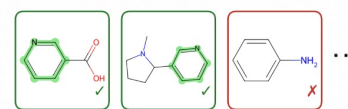
Recompute Mask Using $\mathcal{R}'$
(only reactions allowed by the project)

3. Substructure Constraint (Motif Requirement)

Required Substructure
(USER INPUT)



Candidate Building Blocks
(keep only those that contain the motif)



At the designated step, only sample from blocks that contain the specified motif.

Legend

- Building Block (Token)
- Intermediate / Product
- Data Flow
- RL / Feedback
- Reaction Application
- Constraint / Masking
- Next Token at step t